

What year did *you* come from?: Predicting origin date of book & movie text

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Abstract

Over the years, language has changed dramatically: we do not speak like Shakespearean plays, and even two decades ago, nobody knew what an "oomf" was. This change in language over time is likely reflected in books and movies written at the time. This project attempts to use the language of book text and movie dialogue to predict its year of origin. It uses the Cornell Movie Dialogs Corpus and the Harvard Library Public Domain Corpus and an ordinal regression classifier model on top of a pretrained model to classify text into a decade. The model showed a significant improvement over baselines, and was significantly better at predicting book text than movie text, although on average it was still only within a century of the correct decade. This model shows promising work that could likely be improved upon with more training or improved data, and shows that natural language can in fact be indicative of time of origin.

1 Introduction

Over the years, the usage of language has changed dramatically - people from decades ago have no idea what current slang terms like "oomf" and "slay" mean, and people nowadays do not speak out loud and usually do not write like Shakespeare. This pattern can be spotted in all mediums that natural language is found in: comic books from the Golden Age have a dialogue style that is far different from the modern age, and films and books likely reflect this difference as well. Therefore, this project investigates whether year of origin of natural language is correlated with the content of the language, specifically the decade of origin for book and movie text.

While the problem of predicting date of origin of text has been done before, it has not been exhaustively studied. The related work section will discuss this in more exhaustive detail, but in sum-

mary: there is previous work around dating historical text specifically, but it is few and far between, and mainly focuses on dating historical texts; there is also other work done on related problems such as identifying authors or genres based on book content. None of these previous works deal with mediums other than books, and the works that do investigate date of origin restrict their investigation to strictly historical texts. Therefore, my work charts a different path: investigating whether detecting date of origin from text can be generalized across mediums, and on works ranging from historical to the present day.

To solve this problem, this project uses an ordinal regression model to predict the decade that book text or movie dialogue was written. It uses the Cornell Movie Dialogs Corpus and Harvard Library Public Domain Corpus as its movie and book data sources respectively, and builds an ordinal regression classifier on top of a pretrained distilbert-based uncased model. This model significantly improved upon baselines of choosing a random decade and choosing the most frequent decade. There is ample room for further work, however: the model performed far better on book data than on movie data, and it was on average off by 10 decades.

The problem of determining date of origin has both practical motivations and interesting implications. Determining date of origin of historical texts could help date works whose age has been lost to time, or newly discovered works that are likewise not dated. It is also interesting on its own to track the evolution of language over time and how that shows up in the works we produce. This project has made strides toward accomplishing these goals: it's better than baselines at predicting date of origin, and for the first time establishes that there may be a difference in how natural language changes over time in different mediums (books vs movies), given how there was such a drastic difference in its ability to predict decade for the different medi-

ums. Additionally, with ordinal regression over a pretrained model, baselines were quickly and significantly improved upon, demonstrating that with more data and more training, even better results could be achieved.

2 Data

Data used came from two sources: the Cornell Movie Dialogs Corpus for movie script text, and the Harvard Library Public Domain Corpus for book text.

2.1 Cornell Movie Dialogs Corpus

The Cornell Movie Dialogs Corpus (Danesco-Niculescu-Mizil and Lee, 2011) has 304,713 utterances (lines of dialogue) from 617 different movies. The data is structured into multiple “tables” across .txt files. The relevant ones are movie_lines.txt and movie_titles_metadata.txt.

movie_lines.txt has data formatted as such:

L1045	u0	m0	BIANCA	They do not!
L1044	u2	m0	CAMERON	They do to!
L985	u0	m0	BIANCA	I hope so.

The first three columns are metadata identifiers, and the last one is the actual text.

movie_titles_metadata.txt has data formatted as such:

m0	10 things	1999	6.90	[‘comedy’,]
m1	1492: co	1992	6.20	[‘adventure’,]
m2	15 minutes	2001	6.10	[‘action’,]

The metadata identifier for the movie in the first column links up with the movie_lines.txt column, and the third column provides the year the movie was released.

Therefore, movie dialogue was linked to the year the movie released by joining these two tables. After performing this join, the movie dialogue data had year and line columns.

2.2 Harvard Library Public Domain Corpus

The Harvard Library Public Domain Corpus (Library, 2025) includes 220 billion tokens of text across 230 languages of books from the 1800s and 1900s, and is available on request for educational uses. The data is structured into csv files that list every book available under categories like “NO COPYRIGHT - UNITED STATES” and “IN

COPYRIGHT”. Each entry in the csv files is expanded upon is json format, and there are many json files which hold all the metadata and text of each book. For this project, I used a portion of this data: two json files of in copyright books and two json files of no copyright books, for a total of 2123 books.

Each book’s json entry had many fields, but the relevant ones were as follows:

gxml_date_2	gxml_date_1	text_by_page
1910	19uu	[nLITTAUER LIBRARY
1911	19uu	[1959nALL RIGHTS RE

The first two fields denote possible years that the book was written - due to scanning errors, there are multiple possibilities, hence why sometimes the year shows up with “unknown” characters. The last field contains the text in the book formatted as a list with one page per entry in the list. To process this data, I had to work both the year data and the text data. I processed the year data as such: if both years were valid, take the average; if only one year was valid, choose that one; and if neither year was valid, remove it from the dataset. I then exploded the lists in the text column so that every page of each book formed a new row. After these changes, the book data had year and line columns, the same as the movie data.

2.3 Synthesis

The book and movie data was then joined together, with an extra column named type added to denote whether the data originated from a book or movie script, to enable analysis along those lines. The final data was formatted like this:

year	line	type
1999	Absolutely nothing! That’s t	movie
1908	UP NEW YORK InThe olde	book
1960	Rules of Procedure of the Son	book

There were a total of 746,178 instances, which was split into 80% train, 10% test and 10% val. This is the century class distribution:

Century	1800	1990	2000	9900	5900
Count	20779	422003	65109	196806	41481

Unfortunately, I did not notice that a large portion of the 1900s books had been wrongly scanned as belonging to the 9900th or 5900th century until

after all model training and evaluation was done, and better data validation here would definitely be something to tackle when improving this model in future work. The median decade was 1990.

3 Related Work

The specific niche that this project explores - predicting origin date of not just text, but natural language from other mediums including movie scripts - has not been explored before. However, there is related work that explores similar problems.

3.1 Predicting the Date of Authorship of Historical Texts

This work tackles predicting the origin date of specifically historical text (Tausz, 2011). It uses Project Gutenberg book text as data and attempts to classify it into centuries using three different algorithms: naive Bayes, maximum entropy and least squares, and evaluates the performance of each algorithm. The author finds reasonable success for all three methods, but points out room for further improvement with more data and with analytical methods that take into account syntax. Interestingly, this paper is from 2011, before advances in NLP in the last decade that made it far more easily achievable to have more complex NLP methods to classify text. This is the work most directly related to mine, but it focuses specifically and exclusively on historical texts, and groups text solely into ranges of century, and was also published before recent advances in LLMs. In contrast, my work investigates more than just historical texts and uses buckets that are decade-sized; additionally, it utilizes a model from a 2019 paper, years after this paper was published.

3.2 Author Identification Based on NLP

Identifying not the date of text, but the author, is a question that has been worked on quite a bit. For example, this work (Khalid, 2021) focuses specifically on a news dataset and conducts and compares author identification using several different algorithms, including but not limited to support vector machines, random forest, BERT, and logistic regression. After their experiments, they find that the best accuracy was achieved when the bag of words and latent semantic analysis feature extraction methods were used with BERT. This work differs from mine in that it focuses on news article data and focuses on author identification, but it is

an example of using the style rather than the content or meaning of a text to classify it, and finding reasonable success using a BERT model for this approach, much like I did in this project.

3.3 Book Genre Prediction Using NLP

Classifying books into genres is a thoroughly explored question, and this work (Vayadande et al., 2024) provides an overview of recent research done on it. They analyze different ways that people have tackled the problem of genre classification, and interestingly, find that the aspect of the model that has the most impact on its performance is actually the features it uses; they found that lexical analysis provided useful features for the model to actually train and classify on. This work provides an interesting insight: that the feature-gathering and tokenization phases of creating the model can have a big impact on its performance, and that these phases should be adjusted as per your needs and the task the dataset will be used for. This work isn't directly relevant to this project, but does offer an insight into how future work with data cleaning and carefully picking a tokenizer could possibly improve results - especially given the number of scanning artifacts that could be present within the Harvard Library Public Domain Corpus data.

3.4 Decoding the Past: Explainable Machine Learning Models for Dating Historical Texts

This is a very recent work - preprint published on Nov 28 this year - that is a more recent attempt to tackle the problem of dating specifically historical texts (Pinto et al., 2025). The authors use feature-engineered tree-based ML models for this task and find a 76% accuracy on the century level and a 26% accuracy on a decade level, which was far over baselines. They do find that using the model on a different domain (Project Gutenberg) drops the accuracy significantly, but this still shows success at text dating using non-neural architectures. This work differs from mine in that it uses non-neural architectures and primarily uses feature engineering, and focuses specifically on historical texts, but offers an interesting comparison point to the work I did.

3.5 Estimating the Date of First Publication in a Large-Scale Digital Library

This work focuses specifically on determining *publication* date of books, as opposed to composition

date from text like my work and other work (Bamman et al., 2017). They try two methods: using metadata from records, which can include preprints and multiple editions, and using the content of the text to estimate date. They find that using metadata is best in practice, and that predictions from book content are not as precise; however, estimating from content could still be useful in cases where publication date is unknown or not readily available. This work tackles a different frame of a similar problem to the one my work tackles; one interesting insight from it is that metadata information can at times be more useful than content of text at estimating date. This could extend my work by perhaps using scanned in incomplete dates like 19uu as a starting point for date estimation from the content of the text.

4 Methods

Preparing the data was described in detail in the Data section above. After data from the Cornell Movie Dialogs Corpus and Harvard Library Public Domain Corpus was synthesized, it was split 80/10/10 into training, validation and test sets for use with the model. When the data was loaded into a PyTorch (Paszke et al., 2019) Dataset class, each row of the data was chunked into multiple parts to ensure that parts of the data like the text at the end of the page was not ignored by tokenizer truncation. The data was then cleaned to ensure empty lines were removed, and then formatted so that the years column was changed into decade. The final training dataset had 70,975 instances.

The model itself was an ordinal regression model. Ordinal regression preserves the order of the classes so the model can learn, for example, that the decade 1920 is next to the decade 1930, and thus that a prediction of 1930 would be less wrong than a prediction of 2010, which is far further away. This was chosen because decades are not a set of discrete unordered classes but rather a set of classes that have a set and defined order. This model added one linear layer that acted as an ordinal head on top of a pretrained model. The pretrained model used in this case was distilbert-base-uncased (Sanh et al., 2019). The actual order of the classes was taught to the model through the loss function it used, which was based on the rank-Consistent Ordinal Regression (CORN) paper (Shi et al., 2023). The core idea is thinking of the ranked order as a set of binary classification tasks, where a

single feature of an instance is labeled as true if it exceeds a label, and labeled as false if it less than another label. My loss function modifies standard PyTorch BCEWithLogitsLoss to include this with the forward function:

```
def forward(self, logits, targets):
    targets = targets.view(-1, 1)
    expanded_targets = (targets > torch.arange(
        logits.size(1),
        device=targets.device))
    .float()
    return self.bce(logits, expanded_targets)
```

The model was run in a standard training loop, with the custom loss function and an AdamW optimizer that had a learning rate of 5e-6 set for the pretrained model and a learning rate of 1e-5 set for the ordinal head. The run whose results were used for evaluation ran for 22,000 steps.

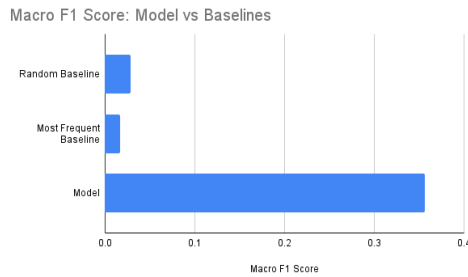
5 Evaluation and Results

Here is the loss curve of the model:

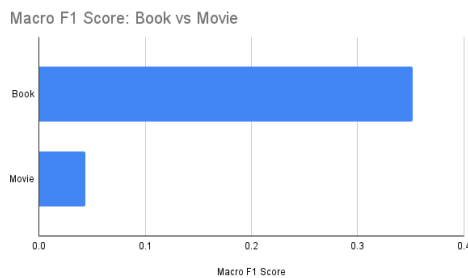


The final trained model was evaluated on a test dataset that was 20,000 randomly selected instances from the test data; data was narrowed to this because of time and computation constraints when evaluating. The main measure used to evaluate the model was macro F1 score, which gives equal importance to every class.

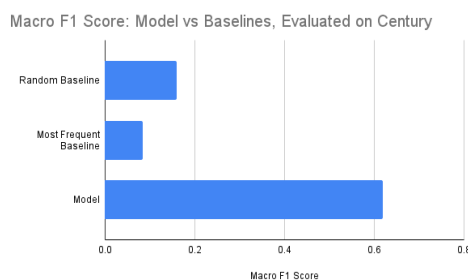
Two baselines were used for evaluation of the model: a "most frequent" baseline, and a "random" baseline. The most frequent baseline was choosing the most frequent label present in the data; this yielded a macro F1 score of 0.01674. The random baseline was choosing a random class; this yielded a macro F1 score of 0.02882. The model's macro F1 score was 0.35620, which was 12.3 times the F1 score of the best performing baseline.



I also evaluated the model's performance on book data versus movie data to see if it performed differently on text from different sources. It did: its macro F1 score for books was 0.35193, whereas its macro F1 score for movies was 0.04335.



Additionally, I evaluated the model's performance over centuries, based on related work that mostly evaluated their own models over that time period. The random baseline over centuries had a macro F1 score of 0.15942, and the most frequent baseline over centuries had a macro F1 score of 0.08370. The model when evaluated by century had a much improved macro F1 score of 0.61913.



One final result is that the mean squared error of the model was 10.5.

6 Discussion

The model performed much better than baselines. In decade and century buckets of time, the model had a 13x or 4x better macro F1 score than the best performing baseline. This is a good preliminary result that shows that going down the route I went down for further work with more and better data, more training, a more robust base pre-trained model, could yield even better results. It also holds promise that the problem of dating text

with NLP with reasonable confidence is one that can be solved.

The model's mean squared error being 10.5 indicates that it was on average by 10 decades, or a century; this is a large error on the decade level, but on the century level, being within one century is a decent preliminary result. This result indicates that although it isn't accurate on average, the model is also not *that* far off on average, which again lends hope to the possibility of more accurately predicting date of origin in future work.

However, there are some results that need to be investigated further. First and foremost, the gulf between the model's performance on book text versus movie text. The model performs remarkably better on book text, and it's not quite apparent why. One explanation is that this is a consequence of me having much more book data available than movie data. Another explanation is that there could be something inherent about movie data that makes it much harder to date: for example, movies are often set in the past (and sometimes the future!) and trying to write past-accurate dialogue could make it harder for the model to detect the difference between origin date of the text and what date the text is trying to sound like it's from; also, movie data here was only dialogue, whereas book data included descriptions around the dialogue and all the other text that is present, so there could be more for the model to go off of in books.

Additionally, the model did not run for as long as I intended it to. It ran for less than an epoch because the job submitted to the Great Lakes GPU ran out of memory; however, I didn't have the time to resubmit the job and garner results and evaluations from a model that was trained for longer, so my results are incomplete.

Another point of contention is the incomplete data cleaning that I discovered when performing evaluations. Some scanned books relaying year of origin as 59xx or 99xx is an oversight that means that decade and century buckets were off, especially since they're ordinally ordered - it means that some 1900s lines that were classified as 59xx or 99xx were ordered to be later than 2000s lines, which could have affected the model significantly. Better data cleaning to solve this problem is definitely something to tackle in future work.

This model is a good preliminary try at solving the problem of detecting date of origin of text with NLP and ordinal regression methods. It performed much better than baselines, which is great and in-

icates that this is a path that can be investigated further. However, looking into why the model performed so much better on book data than movie data, running the model for more epochs with more memory, and cleaning data better by handling invalid years are all ways in which this work could be improved upon.

7 Conclusion

This project tried to show that predicting date of origin of a range of texts - both books and movie scripts - with NLP is a solvable problem. It succeeded: using an ordinal regression model with an ordinal head on top of a pretrained BERT model improved upon baselines when predicting the decade that text originated from by a dramatic amount (13x). It also displayed an impressive macro F1 score on the century level, comparable to recent related work on similar problems. The source code for the project can be found [here](#).

However, there are places to improve this approach: training the model for longer and doing better data cleaning are the two major shortfalls in this model. Additionally, there are ample avenues for future work that expands on this approach. The biggest one is expanding the dataset; the Harvard Library Public Domain Corpus has a mixture of fictional books and things like census reports and research papers in it, and the Cornell Movie Dialogs Corpus has a decent selection of movie dialogue, but there are still many other types of data that could be included for a more thorough look at date of origin determination. For example, incorporating Project Gutenberg books to ensure that more iconic book text was included; and finding databases of TV show scripts, comic book scripts, perhaps music lyrics, personal correspondence, and more. Including these data sources would help round out the goal of this project.

8 Extras

8.1 Other Things I Tried

The thing I spent the most time on that didn't end up being in my final project was actually the other data sources I discussed as directions for future work. I originally started looking for data for this project with the intent of doing it specifically on comic books; I have an interest in superhero comics and am personally acquainted with just how much superhero comic dialogue has changed from the 1970s to now, and I really wanted to find some data

that would help me do this. However, probably due to the combination of comics being a visual medium and a medium where much of the works are in copyright, the best data I could find was a very small dataset of a couple OCR scanned comic pages, so I had to drop this idea. Most of what I tried when creating the actual architecture of the model was included in the final project, though, so there isn't much to talk about in that respect. One note is that I used a smaller BERT model as my pretrained base model because I didn't have the space or time to use a larger one, but I did spend some time trying to get a bigger BERT model to run.

8.2 What I Would Have Done Differently or Next

Many of the things I would have done differently and next are already discussed above in detail, but to list them again: I would have cleaned the year data better, to make sure that the wrongly scanned 59xx and 99xx years were corrected into 19xx for more accurate training data; I would have also re-submitted the job to Great Lakes with a far larger memory and time budget to ensure it could actually train for around 5 epochs; I would have included more sources of data, at the very least including Project Gutenberg data to get some more fictional book data in and hopefully some older books as well. As for things to do next, the biggest one is still finding even more sources of data to generalize this project even more; another avenue that I think would be fun to explore is implementing some of what the very recent related work ([Pinto et al., 2025](#)) explores: using feature engineering to help with date of origin prediction, on top of simple ordinal regression on BERT.

8.3 Group Efforts & Acknowledgements

This project was done entirely alone.

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